Can You Beat DNA Polymerase?

Instructional Manual

Developed by Washington iGEM

1. Overview

In this hands-on activity, students play the role of DNA polymerase — the enzyme used for DNA replication. Each station consists of a wooden block with two vertical dowels. One dowel is stacked with colored beads representing a template strand of DNA, while students race to build the complementary strand on the other dowel. The activity ends with a short video showing how fast the real enzyme really works, and the importance of efficiency towards cell survival.

- **Audience:** Elementary and middle school students
- **Time:** 20–30 minutes per group
- **Group size:** 1–2 students per station. Build 4–8 stations for a typical class rotation.
- **Cost:** roughly $6–8 per station in materials

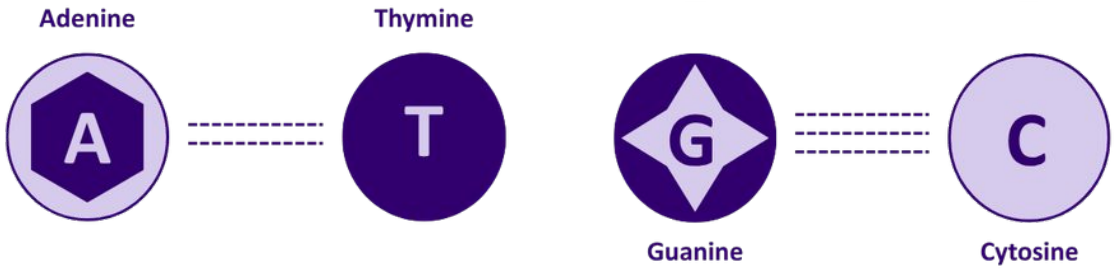
2. Background Information

DNA provides the genetic instructions for every organism on Earth. To store this information, DNA contains 4 chemical bases called nucleotides: adenine (A), thymine (T), guanine (G), and cytosine (C).

DNA is double-stranded, and the two strands are complementary. This means that a nucleotide on one strand only pairs with its partner on the other strand: A pairs with T while G pairs with C. So, if you know the DNA sequence on one strand, you also know the sequence on the other side too!

Every day, the human body produces billions of cells, both to replace dying cells and to allow the body to grow bigger. To achieve this, each cell must replicate all its DNA, so that new cells can also receive a complete set of instructions. The enzyme that does the copying, DNA Polymerase, works by unzipping a DNA strand, reading each base, and then attaching bases that correctly complement the template. Students will repeat this exact process using beads.

To begin the activity, choose 4 bead colors to represent a different nucleotide base (e.g. white as adenine, purple as thymine, etc). Note that there can be many modifications to be colorblind friendly (beads with icons, alternative colors). Adenine pairs to thymine, while guanine pairs to cytosine. Thus, beads designated as adenine should be placed directly across from beads designated as thymine, and so on for the other beads/nucleotides.



3. Supplies

Quantities below are per station (kit). One 48-inch dowel yields four 12-inch strand rods (enough for two kits), and one 8-foot 2×4 yields about fifteen 6-inch blocks.

Materials:

Item	Amount per kit	Estimated cost
Pony beads (At least 0.125in diameter holes, 4 colors)	30+	~ $8 / 160 pcs
Wooden dowel (0.125in diameter, at least 12in height)	2	~ $1 / 48in dowel
2×4 wood, cut into 6in blocks	1	~ $2–4 / 24in board
Containers for loose beads	4	

Tools needed:

Tool	Qty
Saw	1
Drill with 0.125in bit	1
Wood glue	1
Ruler and pencil	1
Sandpaper, medium grit (optional)	1 sheet

Safety Check!

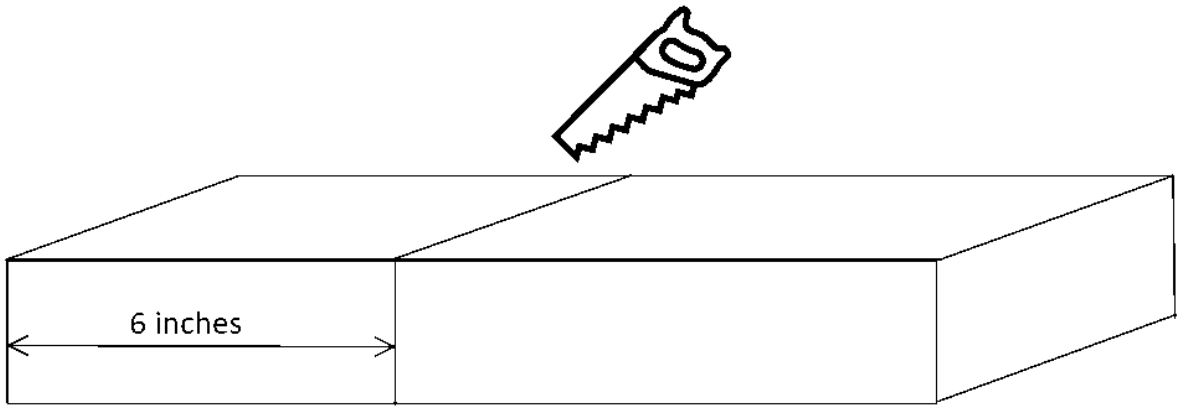
For safety, kits should all be built in advance. Wooden edges should be sanded down to avoid splinters. Be sure to work in a ventilated area!

Beads can be a choking hazard – younger elementary students should be supervised at all times.

4. For Educators: Construction

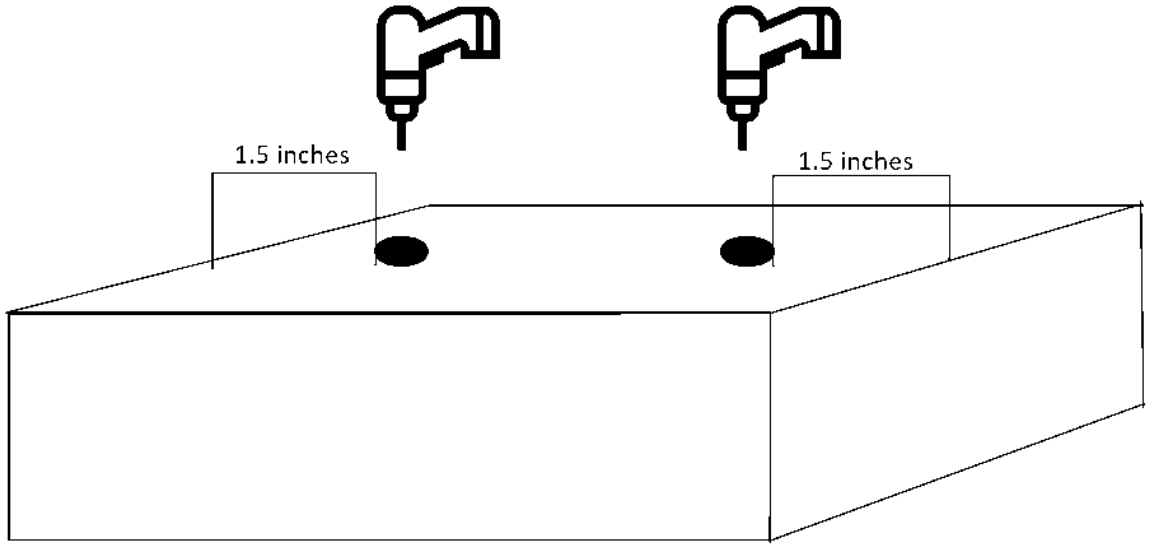
Step 1 — Cut the base block

1. Measure 6 inches from the end of a 2×4 board. Mark a straight line with the ruler and pencil.
2. Cut along the line with the saw, before smoothening the cut edges with sandpaper.



Step 2 — Drill the dowel holes

1. On the top face of the block, mark two points on the center line 1.5 inches from each end. The two marks should be about 3 inches apart.
2. Drill straight down about 0.75 inches deep at each mark.

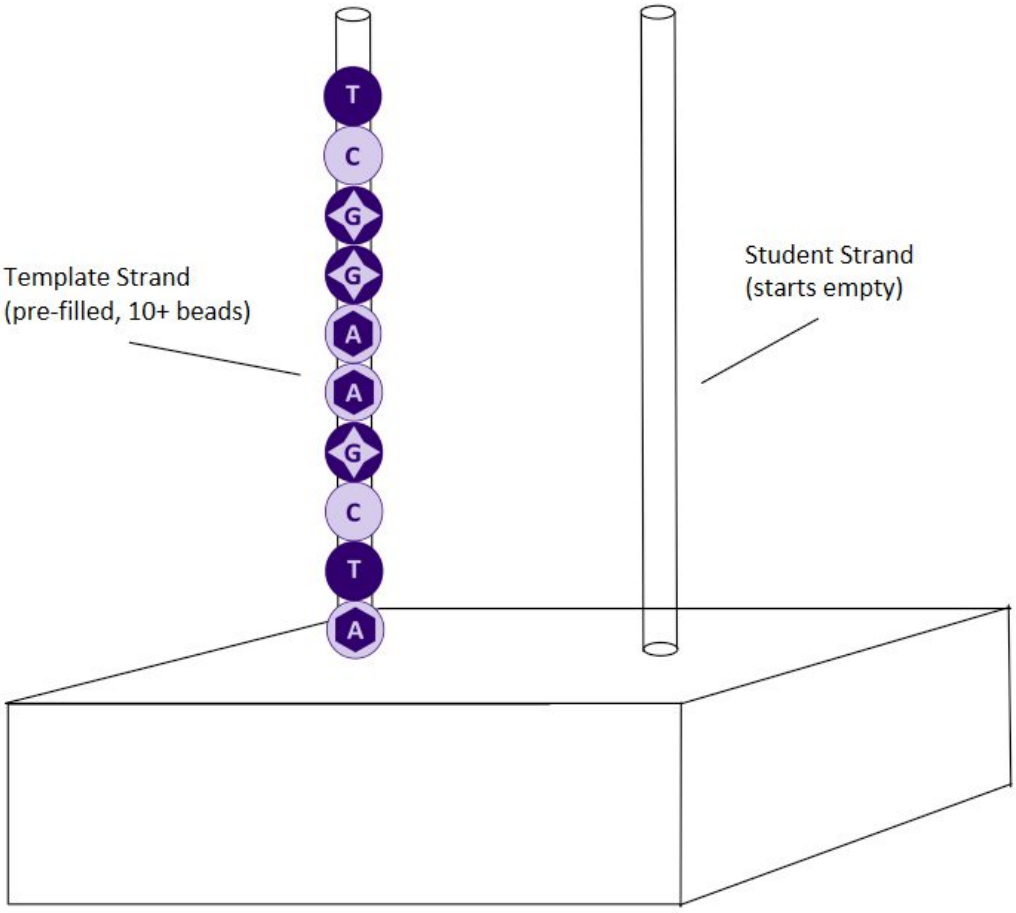


Step 3 — Glue in the dowels

1. If your dowel is longer than 12 inches, cut them into 12-inch segments. Discard any shorter segments.
2. Add one drop of wood glue into each hole.
3. Immediately insert a dowel into each hole and press down firmly for at least 15 seconds. Check that both dowels stand straight, then let cure for at least 15 minutes before handling.

Step 4 — Load the template strand

1. Choose one dowel as the template strand and slide 10+ beads onto it in any order. Be sure to vary the pattern between stations so groups can't copy each other.
2. Leave the second dowel (the student strand) empty.
3. Next to each station, include bowls with different “nucleotides” in each for students to use.
4. **Strongly Recommended:** Write the base letters (A, T, G, C) on a card at each station, or label beads by color using the key in Section 2.



5. Activity Setup

1. Place one finished station in front of each student or each pair of students.
2. Confirm each template dowel holds 10+ beads and each student dowel is empty.
3. Fill a bowl or tray with plenty of loose beads of all four colors, and place it within reach of each station.
4. Post or print the base-pairing color key where every student can see it.
5. Once students complete the activity, play the DNA replication video (link in Section 6) with sound on.

6. Running the Activity

Introduce the science (3–5 minutes)

Briefly introduce DNA as the body's instruction book; every time a cell divides it must copy all of it; the copying machine is an enzyme called DNA polymerase; and it works by matching base pairs — A with T, G with C. Show the color key and check understanding with one or two quick examples ("If the template bead is red, what color goes across from it?").

Run the replication race (10–15 minutes)

1. Send students to their stations and explain the rules: read the template strand from bottom to top, and for each template bead, slide the correct complementary bead onto the empty dowel.
2. Begin the race. The first station to finish with a correct strand wins.
3. Check each finished strand against the template using the color key. A wrong bead is a "mutation", or the insertion of an incorrect DNA base.
4. If time allows, shuffle the template patterns and race again, or have pairs swap stations.

Show the real speed (5 minutes)

Play the DNA replication segment of the enzyme's true speed (2:30-3:20): DNA animation by Drew Berry, WEHI.tv (YouTube). Students will realize that while they could get very efficient, adding nearly 1 bead per second and completing the activity in less than a minute, real human DNA polymerase adds ~50 bases per second. This is because the genome has over 3 billion base pairs, meaning for humans to live, the DNA polymerase must be very efficient.

Wrap up

To ensure students understand the activity, ask the following discussion questions below. Have students return beads into the bowls, reset each student dowel, and prepare for the next rotation.

7. Discussion Questions

- Why does it matter that A only pairs with T, and G only with C?

  It lets one strand act as a template. Each strand carries all the information needed to rebuild the other.

- What would happen if DNA polymerase inserts the wrong base?

  A mutation. In real life, most are harmless, some cause disease, and on rare occasions, some are beneficial.

- When replicating the DNA, did you (students) ever put the wrong base by accident? Were you able to correct the mistake?

  Almost certainly, at least one group accidentally put in the wrong base, but were able to catch the mistake by “proofreading”.

  DNA Polymerase does the exact same thing! Just like how you check the colors to ensure proper matching, DNA Polymerase uses the slightly different shapes of base pairs to check that the pairs fit perfectly.

- Even after checking your work, did some mistakes still happen?

  Yes, mistakes can still happen, even when checking your work! Real DNA polymerase also isn’t perfect, but being such a precise machine, mistakes are very rare.

- Why does the cell need to copy DNA at all?

  To make new cells, every cell needs its own complete instruction manual.

8. Activity Troubleshooting

- **Beads too tight on the dowel?** Purchase thinner dowels, or buy beads with a slightly larger hole. In a pinch, you can also try sanding down the dowel.
- **Dowel wobbles or pops out?** Re-glue with a larger drop, and hold in place for 30 seconds, instead of 15 seconds. For maximum durability, allow wood glue to cure overnight.
- **Up for a challenge?** A fifth colored bead could be used to represent uracil, a nucleotide that replaces thymine in RNA strands. This means that an adenine on a DNA strand pairs with uracil on an RNA strand, while adenine on an RNA strand pairs with thymine on a DNA strand.

  To create a challenge for students, introduce uracil and have the students build up strands of RNA complementary to the template strands of DNA.